



MAE C163A/C263A
Kinematics of Robotic Systems

MATLAB Symbolic Math Toolbox

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Installation

- Open MATLAB
- Click “Add-on”
- Search for “Symbolic Math Toolbox”
- Install

Create Symbolic Variables and Functions

- `syms x`
- `syms x real`
- `syms x integer`
- `assume (x > 1)`

- `syms x(t)`
- `syms f(x, y)`

Symbolic Calculation

- `syms x real`
- `eqn = cos(x)^2 + sin(x)^2`
- `simplify(eqn)`

Differentiation and Integration

- `syms x y real`
- `f(x, y) = x^2 + log(y)`
- `diff(f, x)`
- `int(f, y)`

Solve Equations

- `syms a b c x`
- `eqn = a*x^2 + b*x + c == 0`
- `sol = solve(eqn,x)`
- `nsol = subs(sol,[a b c],[1 2 -1])`
- `vpa(nsol)`

Derive Equations of Motion

- Newton's Method
- Lagrangian Method

Forward Kinematics

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	a_1	0	θ_2
3	0	a_2	0	θ_3
4	0	a_3	0	0

$${}^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_N = {}^0T_1 {}^1T_2 {}^2T_3 \dots {}^{N-1}T_N$$

